

Multi-Agent Reinforcement Learning Based Cooperative Caching With Low Entropy Communications in Fog-RANs

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Abstract—In this paper, we investigate a cooperative edge caching problem in the fog radio access networks (F-RANs). In order to obtain the globally optimal caching strategy that minimizes the content transmission delay and maximizes communication efficiency, we propose a multi-agent reinforcement learning based cooperative caching policy with low entropy communications. First, we propose a double deep Q network (DDQN) based caching policy by taking into account the non-deterministic polynomial hard (NP-hard) aspect of this cooperative caching optimization problem. Then, we extend the state transition model of Markov Decision Process (MDP) under the single agent system into the Stochastic Game (SG) one under the multi-agent system. By employing the DDQN in each agent, the agents can learn and make the global decision for caching. For utilizing the cooperation resources of fog access points (F-APs), the interaction of information is introduced to exchange the historical cache records of cooperative F-APs. However, the information in the interaction may require lower entropy in the fiber link. Therefore, the information entropy is largely reduced to improve the communication efficiency by quantifying the information. Finally, due to the non-computable

gradient of information entropy, we apply a pseudo gradient descent method to approximate the gradient descent in the local model. Simulation results show that our policy achieves better performance in terms of reducing the transmission delay and improving the cooperation among F-APs compared to the benchmark policies. Additionally, it is demonstrated that the proposed policy improves communication efficiency without compromising the performance of cooperative caching.

Index Terms—F-RANs, cooperative caching, multi-agent reinforcement learning, low entropy communications.

I. INTRODUCTION

WITH the rapid development of wireless network technology, the popularity of intelligent terminal devices, and the emergence of a large number of consumer oriented multimedia applications, users frequently connect to wireless networks and generate various content requests, which brings enormous pressure to wireless networks. The continuous emergence of internet services and applications has led to a sharp increase in the demand for data storage and processing. Due to the varying resources required by different applications, these programs typically call for customized solutions [1], [2]. A network architecture called Fog Radio Access Networks (F-RANs), which is closer to the end users, is proposed to address the issue of the forward link's restricted capacity and fully utilize the local cache capacity of edge devices [3]. F-RANs make fog access points (F-APs) located at the edge of the network, which allows some contents to be closer to the users [4]. Edge caching enables these F-APs to store the popular files in a much more local space [5]. Therefore, F-APs can increase the data response rate while reducing the content transmission delay at the meanwhile. How to effectively cache the files has emerged as the main issue with edge caching owing to the limited cache capacities and communication resources [6].

Many research works have now been published on edge caching to find the content replacement policy. In [7], the authors described the cooperative caching problem as a linear integer programming problem (ILP), which was solved by the gradient descent algorithm, and the unbalanced content distribution problem was solved by the Hungarian algorithm. This strategy could drastically lessen redundant data transmission while enhancing the efficiency of information distribution. In [8], the authors proposed a cooperative cache

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placement architecture that transformed optimization problems into local Altruistic games, reducing content transfer time for mobile consumers under multi-users cooperative joint transmission. In [9], the authors divided Base Stations (BSs) into joint cluster sets as cache nodes and proposed a combined cache strategy. The cluster's partial cache space was utilized to store the most recent popular files, with the remaining space used for cooperative caching of less popular contents. In [10], the authors proposed an improved weighted clustering algorithm in a hierarchical architecture to divide edge nodes into clusters, and determined their positions through the cluster heads in the clusters. In [11], the relationship among average interruption probability, BS user density ratio, cache size, and node cluster service radius was derived using the random set method, and the optimal cache distribution was found. In [12], the authors proposed a content recommendation based cooperative caching strategy that is adaptable to changing user requests and allows each F-AP to distributively improve the metrics of fronthaul traffic load and request service time.

Traditional cooperative caching policies, however, generally exhibit poor performance in mobile edge caching due to their limited consideration of specific operating environment characteristics. Factors such as uncertainty in mobile network topology, interference from neighboring devices, dynamic changes in wireless channels, and user mobility significantly impact the effectiveness of caching policies. As a result, traditional approaches fail to adapt to these dynamic conditions, leading to sub-optimal caching performance in mobile edge environments. In recent years, the widespread application of artificial intelligence has transformed various fields. As one of the artificial intelligence technologies, reinforcement learning (RL) has gained significant attention. RL enables an agent to learn and make sequential decisions by actively interacting with the environment. In the dynamic and complex operating environment of mobile edge caching, the optimal caching decision can be made automatically over time by RL, which has the ability to intelligently sense its edge devices. In [13], the cache replacement problem was expressed as an ILP. By taking into account the dynamic characteristics of content popularity and cache decision, it was modeled as a partially observable Markov Decision Process (POMDP), and the optimal caching decision was made by deploying a long short-term memory network (LSTM) to transform the model into a deeply deterministic strategy gradient form. In [14], a Deep Q Network (DQN) was used to determine the cache decision for a single BS. In DQN, a critic framework was generated through a stacked LSTM and a fully connected neural network to evaluate the Q-value, and learning experience was recorded by adding external memory. In [15], the authors focused on cooperative caching strategies for a BS with optimum coherent distance encoding in the ultra dense networks. The Q-learning model was utilized to identify the cache category and content quantity of the BS during off-peak periods. Due to the discretization of model actions, the research also verified the performance of the algorithm in the continuous action. In [16], the authors investigated a Double Deep Q Network (DDQN) based edge caching policy to cope with dynamic user requests with content recommendation in

F-RANs for the sake of maximizing the long-term net profit of each F-AP.

Multi-Agent Reinforcement Learning (MARL), which is an extension of single agent RL and also a promising machine learning architecture, can take into consideration of the interaction and cooperation among multiple agents in an environment. In MARL, each agent had the ability to make decisions, and all individual actions jointly affected the state of the environment. At the same time, because agents may be competitive or cooperative, agents need to consider the effect of joint actions when learning strategies to make globally optimal strategies. In [17], the authors proposed a hierarchical MARL based cooperative caching policy to prevent the local sub-optimal update and improve the influence among other agents. In [18], the authors proposed an algorithm for applying RL into multi-agent system. Each agent had the ability to make decisions, and all individual actions jointly affect the state of the environment. At the same time, because agents may be competitive or cooperative, agents need to consider the effect of joint actions when learning strategies to achieve global optimality. In [19], the authors formulated the cooperative caching problem as an POMDP based multi-agent decision problem and proposed a multi-agent actor critic framework to joint optimize the file transmission costs from the BSs or the cooperative BSs.

Model transition in MARL, however, can also use a significant amount of communication resources. Reference [20]. In order to enhance the cooperation ability between agents, most of works has so far designed a communications framework based on a neural network to realize the information exchanging between agents [21], [22], [23]. As a result, the agents were able to behave collectively rather than independently. Since the agents were incapable to have a global view on the environment, communications became important for performing complex tasks in various scenarios. Especially in the partially observable environments, the communications between agents was crucial technique to coordinate the behavior of each individual. However, most multi-agent communication frameworks only used the cooperation between agents as the only criteria to measure the system performance. In the communication process, the generation and sending of messages were generally arbitrary, which was usually unreasonable under the condition of limited communication resources. In [24], the authors proposed a belief-maintenance procedure in MARL-based cooperative caching policy to observe the historical servers' current strategies. In [25], the authors proposed a deep reinforcement learning (DRL)-based cooperative edge caching approach to enable the distributed edge servers to learn to work together.

Communications efficiency was another significant performance indicator in the F-RANs, which is often ignored by most existing works [26], [27]. Agents conduct joint training and update among agents by transmitting local historical cache information to achieve the optimal cache strategy. Due to the frequent information exchanges among agents in each training round, MARL does not consider the consumption of communication resources caused by the communication process among agents. Reducing the entropy of transmitted

TABLE I
SUMMARY OF MAJOR NOTATIONS

Notation	Description
$\mathcal{N}, \mathcal{N}_n$	Set of F-APs, set of the cooperative F-APs for F-AP n
$\mathcal{U}, \mathcal{U}_n$	Set of users, set of user in the area of F-AP n
$\mathcal{T}$	Total time slots
$\mathcal{F}$	Set of library, set of F contents
$\mathcal{P}_n^t = \{P_{n,1}^t, P_{n,2}^t, \dots, P_{n,f}^t, \dots, P_{n,F}^t\}$	The content popularity distribution of F-AP n
$p_{u,f}^t$	The user's content preference
$\oplus_u$	A random permutation of content library $\mathcal{F}$
$x_{n,f}$	The variable whether F-AP n cache file f
$\mathbf{X}$	The cooperative caching policy
$\mathbf{Y}$	The connectivity among F-APs
$R_{n,u}^t$	The delivery rate between user and F-AP
d_w, d_c, d_b	The transmission delay in F-AP-to-User, F-AP-to-F-AP and Cloud-Server-to-F-AP
S	The cache capacity of F-AP n
$\mathcal{S}_n, \mathcal{S}$	State space of agent n , joint state space
$\mathcal{A}_n, \mathcal{A}$	Action space of agent n , joint state space
$r_n(s_n^t, a_n^t), R(\vec{s}^t, \vec{a}^t)$	Reward function of agent n , global reward function
q_n^t	The user content requesting state of agent n
$Q_n(s_n^t, a_n^t)$	The action value function of agent n
$V_n(s_n^t)$	The state value function of agent n
$\theta_n, \bar{\theta}_n$	The weight parameter of the current Q-network and target Q-network
$S_{-n,f}^t$	The belief of agent n for the content f
$C_{n,f}^t$	The number of times that the requested content has been cached
$\mathcal{D}_n$	The replay memory of agent n
$\vec{m}_{recv,n}^{t-1}$	The received message
$\mathbb{Q}(\cdot)$	The quantization function of the model
$\mathbf{M}$	Set of messages
$H(\mathbf{M})$	The entropy of message

information in interaction may be a effective way to improve the communication efficiency.

Inspired by the above discussions, we proposed a MARL policy with low entropy communications to tackle the issue of cooperative caching in F-RANs. The main contributions can be summarized as:

- 1) We build a content transmission model and propose a content caching policy. By taking into account the time-varying characteristics of user preferences and wireless network channel states, we establish a transmission model for user request content, and then propose an optimization problem to minimize transmission delay to find an optimal caching strategy. The content caching process in a single F-AP is modeled as an MDP.
- 2) We propose a MARL based cooperative caching policy. In order to bring full into the cooperation ability between F-APs, the MDP of a single F-AP is extended to multi-agent system. We deploy DDQN models locally in each F-AP and apply a communication process among the cooperative F-APs. Through the cooperative training of MARL models, an F-AP firstly makes local policy based on the time-varying characteristics of user preferences and wireless network channel states. By interacting with the historical caching information of its cooperative F-AP, the F-AP and its cooperative F-AP jointly update their own local model.
- 3) We study the communication efficiency for the information exchanging among F-APs. In order to reduce

the information entropy of the message, we quantify the messages transmitted in the communication process, and improve the communication efficiency. To overcome the update of F-AP local model under the low entropy communications, we propose a pseudo gradient descent method to solve the problem about the intractable gradient of information entropy. The relevant effectiveness proofs of the pseudo gradient descent method are presented.

- 4) We analyze the transmission performance of the proposed multi-agent reinforcement learning with low entropy communications (MARL-LEC) based cooperative caching policy. Simulation results confirm that our proposed policy can achieve the convergence and demonstrate that the system model parameters of our policy achieve a higher performance compared with considered baseline policies.

The reminder of this paper is organized as follows. Section II introduces the system model and formulates the corresponding optimization problem. Section III describes the proposed cooperative caching policy based on multi-agent reinforcement learning with low entropy communications. Simulation results are shown in Section IV. Final conclusions are drawn in Section V.

II. SYSTEM MODEL

In this section, we firstly introduce the cooperative caching scenario in F-RANs, and then propose the content transmission

model. Finally, we formulate the corresponding optimization problem. For convenience, Table I shows the notations for the important parameters.

A. Network Model

The scenario of cooperative caching in the F-RANs is shown in Fig. 1, which usually consists of a cloud server, F-APs and users. The cloud server is mainly composed of content providers and core networks. An F-AP connects to the cloud server through a backhaul link. The F-AP connects to users via a wireless channel. In addition, fiber links can be used to connect between F-APs. In this cooperative caching scenario, a cloud server can connect to several F-APs, while an F-AP can serve numerous users. In terms of the cooperation mechanism, F-APs in the network synchronize their caching strategies through fronthaul links. This synchronization involves exchanging vital information such as user content request patterns, cache storage statuses, and network traffic conditions. By sharing this data, F-APs can make informed decisions on what content to cache locally, leading to a more efficient use of network resources and an enhanced user experience.

We divide time into discrete time slots, which can be denoted by $\mathcal{T} = \{1, 2, \dots, t, \dots, T\}$. There is one cloud server and N F-APs in this area. Let $\mathcal{N} = \{1, 2, \dots, n, \dots, N\}$ denote the set of F-APs. And let $\mathcal{U} = \{1, 2, \dots, u, \dots, U\}$ denote the set of all users. Moreover, we suppose that the user u_n is exclusively serviced by F-AP n during the time slot t . The set of users in the F-AP n serving cluster can be represented by $\mathcal{U}_n^t = \{1, 2, \dots, u_n, \dots, U_n\}$.

Let $\mathcal{F} = \{1, 2, \dots, f, \dots, F\}$ denote the content library, which is located at the cloud server far away from the users. And the F-APs can access the cloud server to download the contents via backhaul links. Next, we consider the user's content preference in each time slot. Let $p_{u,f}^t$ denote the content preference of user u for content f at time slot t . The content popularity distribution of F-AP n is denoted by $\mathcal{P}_n^t = \{P_{n,1}^t, P_{n,2}^t, \dots, P_{n,f}^t, \dots, P_{n,F}^t\}$, which can be followed by $P_{n,f}^t = \mathbb{E}_u[\sum p_{u,f}^t]$. For the user's content preference, we ignore the impact of user characteristics on it and assume that $p_{u,f}^t$ usually satisfies the M-Zipf distribution [28]:

$$p_{u,f}^t = \frac{\phi_u^t(f)^{-\tau_t}}{\sum_{i=1}^F i^{-\tau_t}}, \quad \forall u \in \mathcal{U}, \quad (1)$$

where τ_t is the skewness factor, and $\phi_u^t(f)$ is a random lists for the content library $\mathcal{F}$ for user u .

The F-APs can cache some contents in each time slot. We assume that every F-AP has the same cache capacity S and that every piece of content has the same size L . The binary variable $x_{n,f}$ indicates if F-AP n has already cached content f . We have $x_{n,f}=1$ if content f has been cached at F-AP n , and otherwise $x_{n,f}=0$. The cooperative caching policy, represented by $\mathbf{X} = [x_{n,f}]_{N \times F}$, should be properly crafted to reduce the content transmission delay for users.

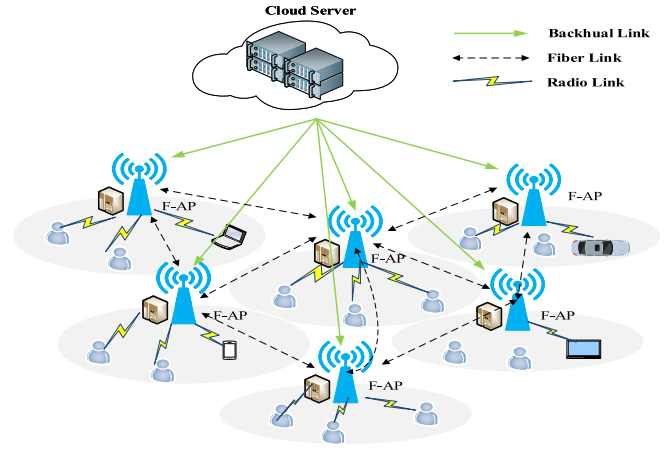


Fig. 1. Illustration of F-RAN.

B. Content Transmission Model

Users can randomly send requests within any time slot. To record the number of requests conveniently, we assume that users will make only one request for the contents in each time slot. When the content f is requested from the user u_n , F-AP n determines how to send the requested content f to user u_n based on its own caching policy $[x_{n,1}, \dots, x_{n,f}, \dots, x_{n,F}]$. The request content is then acquired over the wireless channel through the serving F-AP associated with the user. Assuming that every F-AP has the same bandwidth and transmit power, the delivery rate between F-AP n and user u_n in wireless channel is given by:

$$R_{n,u}^t = B \log \left(1 + |h_{n,u}^t|^2 \frac{P}{N_0 B + P_I} \right), \quad (2)$$

where B is the bandwidth, P is the transmit power of F-AP n , N_0 is the power spectral density of noise, and $h_{n,u}^t$ is the channel coefficient in the wireless channel. For simplicity, we assume that the interference power can be efficiently managed, and thus can be represented by a fixed value P_I ¹ [29]. Therefore, the delay over the wireless channel is given by:

$$d_{w,n,f}^t = \mathbb{E}_u \left[\sum_{u \in \mathcal{U}_n} p_{u,f}^t \frac{L_f}{R_{n,u,f}^t} \right], \quad (3)$$

where L_f is the size of cached file. Let d_w denote the delay for a user to obtain the content from the local serving F-AP, d_c the delay between cooperative F-APs, and d_b the delay of F-AP obtain content from the cloud server. The terms d_w , d_c and d_b are assumed to be fixed values. Let an $N \times N$ matrix $\mathbf{Y}$ denote the connectivity among F-APs, where every binary element $y_{n,m}$ indicates whether F-AP n can be connected with F-AP m . Similarly, $y_{n,m}=1$ if F-AP n can cooperate with F-AP

¹The assumption of constant interference is intended to simplify the analysis and focus on the intrinsic behaviors of the proposed algorithm. It is important to note that while this assumption aids in reducing the complexity of the problem, it may not fully capture the dynamic nature of real-world wireless environments where interference can vary significantly due to factors. This simplification, however, allows us to isolate and evaluate the core functionalities and potential benefits of the proposed algorithm under worst-case conditions.

m , and otherwise $y_{n,m}=0$. Based on the connectivity among F-APs, let $\mathcal{N}_n=\{m|\forall m\in\mathcal{N}, y_{n,m}=1, m\neq n\}$ denote the set of the cooperative F-APs for F-AP n . There are three ways for users to obtain the content:

1) *F-AP-to-User*: The user obtains the content directly from the local F-AP. If the content has been cached in the local serving F-APs, the F-AP can directly transmit the content to the user. For any users' request, the transmission delay can be expressed as:

$$D_{1,n,f}^t = x_{n,f}^t d_{w,n,f}^t. \quad (4)$$

2) *F-AP-to-F-AP*: The users obtains the content from the cooperative F-AP. If the content is not cached in the local serving F-AP, then the F-AP checks whether its cooperative F-APs have cached the content. When the content is cached in a certain cooperative F-AP, the local F-AP can obtain the requested content from its cooperative F-AP and then transmit it to the user. The transmission delay can be expressed as:

$$D_{2,n,f}^t = (1 - x_{n,f}^t) \left(1 - \prod_{m \in \mathcal{N}_n} (1 - x_{m,f}^t) \right) (d_{w,n,f}^t + d_{c,n,f}^t). \quad (5)$$

3) *Cloud-Server-to-F-AP*: The user obtains the content from the cloud server. If the content has not been directly stored in either the local F-AP or the cooperative F-APs, the serving F-AP will obtain the content from the cloud server. The transmission delay is given by:

$$D_{3,n,f}^t = (1 - x_{n,f}^t) \prod_{m \in \mathcal{N}_n} (1 - x_{m,f}^t) (d_{w,n,f}^t + d_{b,n,f}^t). \quad (6)$$

C. Problem Formulation

As discussed above, our goal is to find the globally optimal caching policy $\mathbf{X}^*$, which is the aggregate policy consisted of each F-AP's optimal caching policy. The corresponding cooperative caching problem, which requires minimizing the average transmission delay of the entire system, can be formulated as follows:

$$\min_{x_{n,f}} \bar{D}(\mathbf{X}) = \frac{1}{T} \sum_{t=1}^T \sum_{f=1}^F \sum_{n=1}^N P_{n,f}^t (D_{1,n,f}^t + D_{2,n,f}^t + D_{3,n,f}^t) \quad (7)$$

$$\text{s.t.} \quad \sum_{f=1}^F x_{n,f} L \leq S, \quad \forall n \in \mathcal{N}, \quad (7a)$$

$$x_{n,f} \in \{0, 1\}, \quad \forall n \in \mathcal{N}, \quad \forall f \in \mathcal{F}. \quad (7b)$$

The constraint (7a) implicates that the maximum cache capacity is not exceeded by the size of the cached content S . The constraint (7b) implicates that each content can only cache one in F-AP. It is computationally difficult to directly solve the issue in (7) since we need to determine the best policy for caching the variables $x_{n,f}$ in each time slot. Clearly, the solving such problem is generally NP-hard. The conventional approaches employing model-based algorithms or predetermined rules might be inadequate for determining the best caching strategy under the features of a time-varying

system, because of the dynamic channel conditions and user preference.

III. FRAMEWORK DESIGN OF MARL-LEC

In this section, we transform the content transmission model into an MDP at first. Consider the topology of the F-APs, we then extend MDP to a multi-agent system under the framework of Stochastic Game (SG), and further propose an MARL based cooperative caching policy. Aiming at improving the communication efficiency, we redesign the communication process and reduce the entropy of the information.

A. Markov Decision Process

We use DRL to solve the caching problem in (7). The content transmission model can be modeled as an MDP in the single F-AP. In particular, the agent n performs an action a_n^t in accordance with the observed state s_n^t during time slot t . Then, the agent can be fed back a reward $r_n^t = r(s_n^t, a_n^t)$ from the environment, and then update the new state s_n^{t+1} . The main elements of MDP are denoted as follows:

1) *State Space*: In F-RANs, the agent's state space can be divided into the content caching state of F-AP and the content requesting state. In time slot t , the content caching state is denoted by $\mathbf{x}_n^t = \{x_{n,1}^t, \dots, x_{n,f}^t, \dots, x_{n,F}^t\}$. The content requesting state can be denoted by $\mathbf{q}_n^t = \{f|f \in \mathcal{F}\}$, which is the set of F-AP n content requesting in time slot t . Overall, the state s^t is represented by the two vectors as follows:

$$s_n^t = \{\mathbf{x}_n^t, \mathbf{q}_n^t\}, \quad s_n^t \in \mathcal{S}_n, \quad (8)$$

where the state space $\mathcal{S}_n$ is a collection of all states in the agent n .

2) *Action Space*: An F-AP needs to decide what content requested by the users can be cached (in a local F-AP, a nearby cooperative F-AP, or a cloud server), and what content needs to be replaced when the F-AP's cache capacity reaches its maximum. Thus, the selected action a^t of agent n can be expressed as:

$$a_n^t = \{\mathbf{a}_n^t\}, \quad a_n^t \in \mathcal{A}_n, \quad (9)$$

where $\mathbf{a}_n^t = [a_{n,0}^t, a_{n,1}^t, \dots, a_{n,s}^t, \dots, a_{n,S}^t]$ is a cache replacement update action, and $a_{n,0}^t \in \{0, 1\}$ indicates that the contents of F-AP n cached will not be replaced, $a_{n,s}^t=0, s \in S$ indicates that the s th cached content of F-AP n in the cache space will not be replaced, $a_{n,s}^t=1, s \in S$ indicates that the s th content cached by F-AP n in the cache space will be substituted by the content requested by the current user. The action space $\mathcal{A}$ is a collection of all selectable actions.

3) *Reward Function*: In the multi-agent system, the training goal is to optimize the joint strategy of the agents for F-AP content caching. Therefore, the reward function for each agent is expressed as:

$$r_n(s_n^t, a_n^t) = \prod_{f=1}^F e^{-P_{n,f}^t (\lambda_1 D_{1,n,f}^t + \lambda_2 D_{2,n,f}^t + \lambda_3 D_{3,n,f}^t)}. \quad (10)$$

where λ_1, λ_2 and λ_3 denote the weighting factors, and $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

B. Stochastic Game Process

In MDP, the agents can learn how to maximize the reward during the discrete time $t = 0, 1, \dots, T$ by interacting with the environment. However, the cooperation among F-APs is not brought fully into and agents could interact with other agents to jointly make the optimal caching policy. Thus, we extend MDP into the multi-agent system. Consider an single F-AP as an agent in F-RAN, the agents execute actions simultaneously, and the rewards of agents are generally arbitrarily related. Tuple $\langle N, \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$ can be defined as an SG model with N agents, where $\mathcal{S} = \mathcal{S}_1 \times \dots \times \mathcal{S}_N$ is the joint state space for all agents, each agent state $s_n^t \in \mathcal{S}_n$ in SG consists of the individual states $\vec{s}^t \in \mathcal{S}$ of all agents during time slot t ; $\mathcal{A} = \mathcal{A}_1 \times \dots \times \mathcal{A}_N$ is the joint action space of all agents, the current joint of all agents can be denoted by $\vec{a}^t = (a_1^t, \dots, a_n^t, \dots, a_N^t) \in \mathcal{A}$; $P: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the state transition probability mapping, the system select the joint action $\vec{a}^t$ to update the joint state from $\vec{s}^t$ to $\vec{s}^{t+1}$ with the probability $P_{\vec{s}^t \vec{s}^{t+1}}^{\vec{a}^t}$, which satisfy $\sum_{\vec{s}^{t+1}} P_{\vec{s}^t \vec{s}^{t+1}}^{\vec{a}^t} = 1$; $R: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the global reward function of all agents, which evaluates the quality of state transition. Each agent n performs action selection based on its own strategy π_n , and the joint strategy of all agents can be denoted by $\pi = \{\pi_1, \dots, \pi_N\}$. In practice, agents in the SG always operate as a group, thus the following assumptions are acceptable and serve as the learning method's foundations.

Assumption 1: Agents have mutual interests. The rewards for the agents in SG are taken from the same distribution. Based on the definition of reward function for each agent in the Eq. (11), the global reward function is expressed as:

$$R(\vec{s}^t, \vec{a}^t) = \sum_{n=1}^N \pi_n(a_n^t | s_n^t) r_n(s_n^t, a_n^t). \quad (11)$$

Assumption 2: Agents may always see the others' state transitions as a group, implying that the SG's combined state is observable.

The Markov property also applies to state transitions, which states that the probability distribution cannot be predicted based on a sequence of events that have led to the present state. A reward and the next state can only be determined by the current state and the joint actions of all agents [30].

C. MARL: A Multi-Agent Reinforcement Learning Based Cooperative Caching Policy

Each agent is expected to find the optimal strategy π_n to maximize long-term reward, and the long-term cumulative reward of agent n , as well as state value function, can be expressed as:

$$V_n(s_n^t) = \mathbb{E} \left[\sum_{\tau=t}^{\infty} \gamma^{\tau-t} R(\vec{s}^{\tau}, \vec{a}^{\tau}) | s_n^t \right] = R(\vec{s}^t, \vec{a}^t | s_n^t) + \gamma \sum_{\vec{s}^{t+1}} P_{\vec{s}^t \vec{s}^{t+1}}^{\vec{a}^t} V_n(s_n^{t+1}), \quad (12)$$

where $\gamma \in [0, 1]$ is the discounted factor that reflects the importance attached to the future reward, and $P_{\vec{s}^t \vec{s}^{t+1}}^{\vec{a}^t}$ is the transition. The optimal strategy of agent n for maximizing the

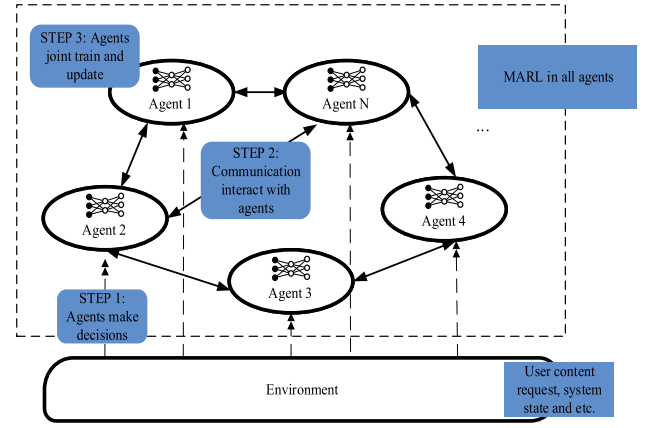


Fig. 2. Illustration of the scenario in F-RAN.

rewards can be expressed as:

$$\pi_n^*(s_n^t, a_n^t) = \arg \max_{\pi_n} V_n(s_n^t). \quad (13)$$

The optimal strategy π_n^* is the best response to its cooperative agents. $\{\pi_1^*, \dots, \pi_n^*, \dots, \pi_N^*\}$ can be considered as the global optimal strategy.

The value of which the action a_n^t is executed by agent n under the current state s_n^t can be represented by the action value function $Q_n(s_n^t, a_n^t)$, and the action value function is the specific representation of state value function when executing an action. The action value function can be expressed as:

$$Q_n(s_n^t, a_n^t) = R(\vec{s}^t, \vec{a}^t | s_n^t, a_n^t) + \gamma \sum_{\vec{s}^{t+1}} P_{\vec{s}^t \vec{s}^{t+1}}^{\vec{a}^t} V_n(s_n^{t+1}). \quad (14)$$

In Q-learning, the update of action value function can be expressed as:

$$Q_n(s_n^t, a_n^t) \leftarrow (1 - \alpha) Q_n(s_n^t, a_n^t) + \alpha [R(\vec{s}^t, \vec{a}^t | s_n^t, a_n^t) + \gamma \max_{a_n^{t+1}} Q_n(s_n^{t+1}, a_n^{t+1})], \quad (15)$$

where α is the network learning rate to indicate how far the current action value $Q_n(s_n^t, a_n^t)$ is adjusted towards to the $R(\vec{s}^t, \vec{a}^t | s_n^t, a_n^t) + \gamma \max_{a_n^{t+1}} Q_n(s_n^{t+1}, a_n^{t+1})$.

1) *Double Deep Q Network:* Conventional Q-learning methods are inefficient for high-dimensional state and action spaces due to greedy strategies are required to find state-action pairs. Therefore, using a DQN based algorithm that combines Q-learning and deep neural networks to estimate the action value function is much more efficient approach. In addition, algorithms based on DQN may overfitting the action value function, resulting in significant deviations in the model. To overcome the curse of dimensionality and make the optimal action to maximize the reward, RL techniques including DQN and DDQN are utilized [31]. In comparison to the traditional DQN algorithms, the DDQN introduces a critical improvement in the learning process by decoupling the action selection from the evaluation of its value. This distinction is achieved through the use of two separate neural network architectures within DDQN: a current Q-network and a target Q-network. The current Q-network is responsible for the selection of the action, while the target Q-network independently evaluates the value

of that action. This separation effectively mitigates the issue of value overestimation inherent in DQN [32]. By preventing the simultaneous use of the same network for both action selection and value estimation, DDQN enhances the accuracy and stability of the learning process. This architecture makes DDQN particularly suited for identifying the optimal policy in environments where overestimation can significantly impact the performance. The current Q-network $Q_n^t(s_n^{t+1}, a_n^{t+1}|\theta_n)$ is used to make approximations $Q_n^t(s_n^t, a_n^t)$ in Eq. (14), where θ_n is the weight parameter of the current Q-network. The target Q-network $\hat{Q}_n^t(s_n^t, a_n^t|\hat{\theta}_n)$ is applied in order to calculate the target Q-value, $\hat{\theta}_n$ is the weight parameter of the target Q-network. And the Q-value of target Q-network can be expressed as follows:

$$\hat{Q}_n(s_n^t, a_n^t|\hat{\theta}_n) = R(s_n^t, a_n^t) + \gamma \max_{a_n} \hat{Q}_n(s_n^{t+1}, a_n^{t+1}|\hat{\theta}_n), \quad (16)$$

where $a = \arg\max_{a_n} Q_n^t(s_n^{t+1}, a_n^{t+1}|\theta)$ is an action made from the current Q-network to keep the current Q-value under the state s_n^{t+1} , and $\hat{\theta}_n$ is the target Q-network parameter.

By copying the network parameters of the target Q-network from the current Q-network at regular intervals rather than updating them continually, delay update lessens the correlation between the target Q-value and the current Q-value. The loss function is updated using a gradient descent method [33], as follows:

$$f_n(\theta_n) = \mathbb{E}[(\hat{Q}_n(s_n^t, a_n^t|\hat{\theta}_n) - Q(s_n^t, a_n^t|\theta_n))^2] + \mathbb{E}[\nabla[\hat{Q}_n(s_n^t, a_n^t|\hat{\theta}_n)]], \quad (17)$$

where the target Q-network parameter $\hat{\theta}_n$ will copy θ_n from the current Q-network $Q_n^t(s_n^t, a_n^t|\theta_n)$ every ν steps. Among them, the last term of the Eq. (17) is the variance of the target Q-network, which is not dependent on the weight parameter θ of the current Q-network and can usually be ignored. Using the Stochastic Gradient Descent (SGD) [34] to update the weighting parameters of the current Q-network, the gradient can be expressed as:

$$\nabla_{\theta_n} f_n(\theta_n) = \mathbb{E}[(\hat{Q}_n(s_n^t, a_n^t|\hat{\theta}_n) - Q_n(s_n^t, a_n^t|\theta_n)) \nabla_{\theta_n} Q_n(s_n^t, a_n^t)]. \quad (18)$$

2) *Frequency Maximum Q-Value*: Q-learning always converges on the best course of action for single-agent setup, regardless of the strategy used to choose that course of action. The action sampling technique, however, becomes essential for learning convergence in the multi-agent scenario. It is difficult to develop a strategy for selecting actions that strikes a balance between investigating the utility of moves that have been tried a few times and exploiting those in which the agent is confident of receiving a large reward [17].

Since the global reward depends on all agents, each agent needs to observe the historical caching information of other agents to adjust the local caching policy. MARL introduces communication processes to achieve information exchanging between agents. The agent $n \in \mathcal{N}$ is associated with the current cache policy of its cooperative agent $m \in \mathcal{N}$ when caching files. In order to measure the local caching policies of other agents, each agent observes the historical caching

policies of other agents and maintains confidence in their historical caching strategies. We consider that the observation value of cooperative agent $m \in \mathcal{N}$ can be denoted as an indicator of the agent n 's caching strategy. Using $C_{n,f}^t$ represents the number of times the requested content f has been cached before the time slot t (the content f is cached locally until it is replaced by other content and marked as cached once). The agent n will take the action a based on its selection to record the communication indication $C_{n,f}^t$ during the current time slot t , and the belief of historical caching information can be expressed as:

$$S_{-n,f}^t = \prod_{\substack{x_{m,f}=1 \\ m \in \mathcal{N}_n}} \left(\frac{C_{m,f}^t}{t} \right) \prod_{\substack{x_{m,f} \neq 1 \\ m \in \mathcal{N}_n}} \left(1 - \frac{C_{m,f}^t}{t} \right). \quad (19)$$

We therefore consider the Frequency Maximum Q-value (FMQ) method. Instead of updating the action value function in Eq. (16), we rewrite this function as follows:

$$\hat{Q}_n^t(s_n^t, a_n^t|\hat{\theta}_n) = \hat{Q}_n^t(s_n^t, a_n^t|\hat{\theta}_n) + S_{-n,f}^t(R(s, a) + \gamma \max_{a_n} \hat{Q}_n^t(s_n^t, a_n^t|\hat{\theta}_n)) \quad (20)$$

Agents cooperate with their cooperative agents to update their local models. In each round of time slots, a agent n will decide the action and store the records $[s_n^t, a_n^t, R(s^t, a^t), s_n^{t+1}, S_{-n,f}^t]$ in the agent n 's experience replay memory $\mathcal{D}_n$. The agent will randomly extract a batch of data from the experience replay pool for local network updates. The loss function of the local model can be rewritten as:

$$f_n(\theta_n) = (\hat{Q}_n(s^t, a^t|\hat{\theta}_n) - Q_n(s^t, a^t|\theta_n))^2. \quad (21)$$

The training objective is to minimize the redefined loss function in order to update these parameters:

$$\nabla_{\theta_n} f_n(\theta_n) = (\hat{Q}_n(s^t, a^t|\hat{\theta}_n) - Q_n(s^t, a^t|\theta_n)) \nabla_{\theta_n} Q_n(s^t, a^t|\theta_n). \quad (22)$$

In each training round, every agent interacts with its cooperative agent and records the historical caching belief of cooperative agent. Then, agent updates the local model by the loss function and make the optimal caching strategy. More details of MARL are outlined in Algorithm 1. The parameter θ of DDQN framework is updated by minimizing the loss function in Eq. (17). The effectiveness of SGD can demonstrate the convergence of MARL. The effectiveness of the MARL policy is then proved by the following theorem.

Theorem 1: Based on the local model of loss function f_n , the corresponding loss function can be expressed as:

$$F(\theta^t) = \frac{1}{N} \sum_{n=1}^N f_n(\theta_n^t). \quad (23)$$

Taking the numerical convexity of the loss function into account, the gradient of loss function satisfies Lipschitz continuity under the constant G . For θ^t and θ^{t+1} , we can get:

$$\|\nabla F(\theta^t) - \nabla F(\theta^{t+1})\| \leq G \|\theta^t - \theta^{t+1}\|. \quad (24)$$

Proof: Please see Appendix A. ■

Algorithm 1 Proposed MARL Based Cooperative Caching Policy

Require: The parameter $\alpha, \gamma, \lambda_1, \lambda_2, \lambda_3, \tau_t, d_w \ll d_c < d_b$;
Ensure: $\mathbf{X}^*$;

```

1: Initialize the replay memories  $\mathcal{D}_1, \dots, \mathcal{D}_N$ , the current Q-
   network  $Q$  with the weight  $\theta$  and the target Q-network  $\hat{Q}$ 
   with the weight  $\hat{\theta} = \theta$ ;
2: Set the historical records about the contents to 0 and set
    $C_{n,f} = 0$ ;
3: for time slot  $t = 1$  to  $T$  do
4:   for F-AP  $n = 1$  to  $N$  do
5:     User  $u_n$  requests the content  $f$ ;
6:     if  $x_{n,f} = 1$  then
7:       Delivery the content  $f$  to the user  $u$ ;
8:       End episode;
9:     else
10:      Perform the action  $a_n^t = \arg\max_a Q_n^t(s, a)$ ;
11:      Update the historical records  $C_{n,f} = C_{n,f} + 1$  and
         $C_{n,a_n^t} = 0$ ;
12:      Execute the selected action  $a_n^t$ ;
13:      Update the  $a_n^t$ -th stored file in F-AP  $n$  with the
        content  $f$ , and get the new state  $s_n^{t+1}$ ;
14:    end if
15:    Compute the belief  $S_{-n,f}^t$  based on (19);
16:    Save the experience  $[s_n^t, a_n^t, R^t, s_n^{t+1}, S_{-n,f}^t]$  in  $\mathcal{D}_n$ ;
17:    Update the state  $s_n^t = s_n^{t+1}$ ;
18:    for F-AP  $m \in \mathcal{N}_n \cup \{n\}$  do
19:      Calculate the reward  $R^t$  based on (11);
20:      Randomly sample experiences
         $[s_m, a_m, R, s_m, S_{-m,f}]$  from local replay
        memory  $\mathcal{D}_m$ ;
21:      Update the target Q-value  $\hat{Q}_m^t(s_m^t, a_m^t | \hat{\theta}_m)$  based
        on (20);
22:      Execute a gradient descent step to update the
        weight  $\theta_m$  based on (22);
23:      Replicate  $\hat{\theta}_m = \theta_m$  every  $D$  time slots;
24:    end for
25:  end for
26: end for

```

Theorem 1 demonstrates the effectiveness of the MARL policy. In summary, the MARL policy can apply a communication process to enhance the cooperation of F-APs. By interacting the historical caching information, F-APs can make the globally optimal cache policy.

Remark 1 To enhance the adaptability of our reinforcement learning model to dynamic network conditions, strategies such as transfer learning are recommended. Transfer learning allows models to leverage knowledge from previously learned tasks to new but related problems efficiently. A relevant example of this in network management is illustrated by Xiang et al. [35], who employed a deep reinforcement learning algorithm to optimize content caching and mode selection in fog-RAN slicing. Their approach dynamically adapts to changing channel states and cache statuses, significantly enhancing performance metrics such as hit ratio and sum transmit rate [35]. Additionally, incremental learning could be explored, which allows continuous model updates as new data becomes available, ensuring the model remains effective over time. Implementing these adaptive learning strategies will enhance the robustness and applicability of our model under variable network conditions [36].

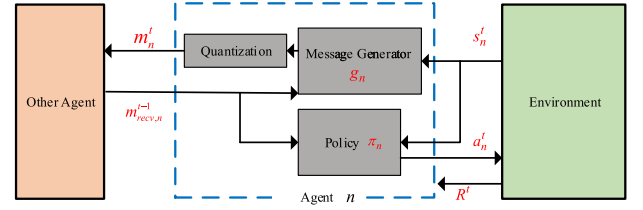


Fig. 3. The communication process of agent n .

D. MARL-LEC: A MARL With Low Entropy Communication Based Caching Policy

An MARL framework is proposed, in which agents perform joint training and update between agents by transmitting local historical cache information to achieve the optimal cache strategy. Due to frequent information exchange between agents in each training round, this algorithm does not consider the consumption of communication resources caused by the communication process between agents. Due to the existence of two types of data transmission between F-APs: content transmission and interaction of historical caching information, it consumes a huge amount of communication resources. In F-RANs, the research direction of improving the communication efficiency between agents in multi-agent system is mainly divided into two directions: one is communication object scheduling, including selecting communication timing and communication target, that is to say, to balance agents, only some agents are selected to communicate in each time slot. However, this policy generally requires a core agent to collect the weights of all agents to evaluate the importance of agents; the second is to reduce the information entropy of generated messages. According to Shannon Hartley's theorem [37], the lower the entropy of messages, the less bandwidth is required. Therefore, messages with low entropy can be reliably transmitted on poor communication channels. Thus, the goal is to improve the communication efficiency among agents by reducing the information entropy of messages on the premise of ensuring the performance of MARL algorithm does not decline.

In the system, the agent n requires a policy π_N , message generator g_n to complete the cache policy update. The process of an agent interacting with the environment and other agents complete within a time slot. In the time slot t , the agent n first obtains the current state S_n^t from the environment and accepts messages from cooperative agents $\vec{m}_{recv,n}^{t-1}$. The received message is expressed as:

$$\vec{m}_{recv,n}^{t-1} = \{\vec{m}_1^{t-1}, \dots, \vec{m}_{n-1}^{t-1}, \vec{m}_{n+1}^{t-1}, \dots, \vec{m}_N^{t-1}\}. \quad (25)$$

The agent selects an action $a_n^t = \pi(s_n^t, \vec{m}_{recv,n}^{t-1})$ based on the current observations and the received messages, and then broadcasts its own message to other agents. Finally, the environment provides feedback to the agent n for the currently selected reward R_n^t . In our system, the messages generated by the agent are first quantified before being broadcast to other agents. We set the output range of the agent message generator to $[-1, 1]$, and define the integer quantization function as:

$$\mathbb{Q}(x) = k\Delta - 1, \quad \forall x \in [(k - 0.5)\Delta - 1, (k + 0.5)\Delta - 1], \quad (26)$$

where Δ is the quantization interval length, $k \in \{0, 1, \dots, K\}$, $K=2/\Delta$ is the number of quantization intervals. We assume that message transmission between agents is harmless, our goal is to reduce the entropy of the message source, which is independent of the channel state.

To calculate the entropy of quantified messages, we first define a binary function as:

$$I_k(x) = \begin{cases} 1, & x \in [(k-0.5)\Delta - 1, (k+0.5)\Delta - 1), \\ 0, & x \notin [(k-0.5)\Delta - 1, (k+0.5)\Delta - 1). \end{cases} \quad (27)$$

Firstly, consider setting the length of the message to 1, which indicates that a message is a number rather than a vector. For a given message set $\mathbf{M} = \{m_1, m_2, \dots, m_N\}$, the entropy of quantified message is given by:

$$H(\mathbf{M}) = - \sum_{k=0}^K \frac{\epsilon + \sum_{n=1}^N I_k(m_n)}{N} \log \frac{\epsilon + \sum_{n=1}^N I_k(m_n)}{N}, \quad (28)$$

where ϵ is a fixed polar decimal to avoid the occurrence of $\log 0$ in the calculation. When the length of a message exceeds 1, the entropy of message can be calculated by summing the entropy of each bit of the message.

We aim to reduce message entropy in an MARL framework with communication processes without compromising the performance of cooperative caching. The method that we adopted is to add a regularization [38] to the MARL training target in Eq. (21). The adjusted training function is as follows:

$$f'_n(\theta_n) = f_n(\theta_n) + \beta H(\mathbf{M}_n), \quad (29)$$

where β is the weight of the regularizer and its value depends on the trade-off between minimizing $f_n(\theta_n)$ and minimizing $H(\mathbf{M}_n)$. Based on the new training function in (29), the gradient of the network model parameters is adjusted to:

$$\nabla_{\theta_n} f'_n(\theta_n) = \nabla_{\theta_n} f_n(\theta_n) + \beta \nabla_{\theta_n} H(\mathbf{M}_n). \quad (30)$$

Because of $I_k(\cdot)$ has a gradient of 0 or ∞ , $\nabla_{\theta_n} H(\mathbf{M}_n)$ is also 0 or ∞ , which results in a SGD-based process that cannot be calculated (30). To reduce the information entropy, we use Pseudo Gradient Descent (PGD) to solve this problem. Information entropy $H(\mathbf{M})$ can be seen as N multivariate function $H(m_1, m_2, \dots, m_N)$. We focus on how to adjust m_n to reduce $H(\mathbf{M})$.

We use the function $p_k(m_i)$ instead of gradient $\nabla_{m_n} I_k$:

$$p_k(x) = \begin{cases} 1, & x \in ((k-1)\Delta - 1, k\Delta - 1) \\ -1, & x \in (k\Delta - 1, (k+1)\Delta - 1) \\ 0, & x = k\Delta - 1, \end{cases} \quad (31)$$

Furthermore, the pseudo gradient can be expressed as:

$$\nabla_{m_n}^p H(\mathbf{M}) = \sum_k \nabla_{I_k} H(\mathbf{M}) p_k(m_n). \quad (32)$$

Next, for the update of the parameter in local model, we can establish the following theorem.

Theorem 2: Since $\nabla_{m_n} H(\mathbf{M}_n) = \sum_k \nabla_{I_k} H(\mathbf{M})$, $\nabla_{m_n} I_k = 0$ or ∞ , we use the pseudo gradient $\nabla_{m_n}^p H(\mathbf{M})$ can achieve:

$$\eta \rightarrow 0, m'_n = m_n - \eta \nabla_{m_n}^p H(\mathbf{M}), H(m_1, \dots, m'_n, \dots, m_N) \leq H(m_1, \dots, m_n, \dots, m_N). \quad (33)$$

Proof: Please see Appendix B. ■

Theorem 2 demonstrates the pseudo gradient method has the property of gradient descent and achieve the low entropy communication. Based on the above analysis, in each iteration, the parameters θ_n of the current Q network can be determined by using a random gradient descent for the state value function combined with the PGD method of information entropy. Then, the parameter θ_n can be updated by:

$$\theta_n = \theta_n - \eta (\nabla f_n(\theta_n) + \beta \sum_k \nabla H(\mathbf{M}) p_k(m_n)). \quad (34)$$

The procedure of MARL-LEC is presented in Algorithm 2.

Algorithm 2 The MARL-LEC Based Cooperative Caching Policy

- 1: Initialize the replay memory $\mathcal{D}_n$, the current Q-network Q with the weight θ_n , and the target Q-network $\hat{Q}$ with the weight $\hat{\theta}_n = \theta_n$;
- 2: Set the relevant messages of F-APs to 0;
- 3: **for** time slot $t = 1, 2, \dots, T$ **do**
- 4: Initialize the message body $\mathbf{M} = (\vec{0}, \dots, \vec{0})$;
- 5: **for** F-AP $n = 1, 2, \dots, N$ **do**
- 6: Agent n receive the message $\vec{m}_{recv,n}^{t-1}$;
- 7: Observe the state $\mathbf{s}_n^t = \{q_{n,1}^t, q_{n,2}^t, \dots, q_{n,S}^t, f\}$ and choose an action a_n^t ;
- 8: Compute the local message m_n and get the quantified message according to (26);
- 9: Update the message $\mathbf{M}$;
- 10: Compute the information entropy $H(\mathbf{M})$ according to (28);
- 11: Agent n transfer the message m_n to the cooperative agent $m \in \mathcal{N}_n$;
- 12: Update the local parameter θ_n according to (34);
- 13: **end for**
- 14: **end for**

E. Extension to Inconstant Interference Case

In our current model, we assume constant interference to simplify the analysis and focus on evaluating the core functionalities and benefits of the proposed multi-agent reinforcement learning (MARL) algorithm. The constant interference assumption allows us to isolate the effects of the MARL algorithm on the cooperative caching strategy without the added complexity of dynamically changing interference conditions. This simplification provides a controlled environment to validate the fundamental behavior of the algorithm, ensuring clarity in interpreting the results. However, this assumption also implies that the model may not fully capture the variability of real-world wireless environments, where interference can fluctuate due to factors such as user mobility, physical obstructions, and varying traffic loads.

To enhance the practical applicability of our framework, we propose extending the current model to account for variable

interference. This can be achieved by incorporating stochastic elements that reflect the dynamic nature of interference in real-world scenarios. Specifically, we can model interference as a time-varying random process influenced by long-term channel fading (shadowing) and short-term variations (fast-fading). The proposed extension involves updating the state and reward functions in the MARL algorithm to include interference as a variable component. This dynamic interference model will enable the agents to learn and adapt to varying interference conditions over time. Techniques such as transfer learning and incremental learning, as discussed by Xiang et al. [35] in their work on fog-RAN slicing, can be employed to periodically update the model parameters based on observed environmental changes, ensuring sustained performance and adaptability in dynamic network conditions.

Transfer learning can be used to adapt MARL models more effectively to the cooperative caching problem as interference fluctuates over time. Instead of training the model from scratch for each new scenario, it can transfer learned policies from previous phases with similar caching and interference dynamics, reducing exploration time and speeding up convergence. For instance, when interference varies due to user mobility or traffic changes, agents can reuse effective policies from similar past scenarios and fine-tune them to current conditions, thereby reducing learning time and maintaining optimal performance, especially in predictable environments.

IV. SIMULATION RESULTS

The performance of the proposed MARL-LEC cooperative caching policy is evaluated via simulations. We suppose that the content library has 1000 contents and that each content is 1 MB in size. The small-scale channel gain $|h_{n,u}^t|^2$ follows independently standard exponential distribution in each time slot [29]. The setting of other parameters are given in Table II. In the hidden layer of the DDQN, a fully connected feedforward neural network is employed to build both the current Q-network and the target Q-network.

We adopt the following caching schemes as benchmarks.

- 1) **Least recently used (LRU)**: The stored content is that the most recently requested stored content will be replaced by the new content.
- 2) **Greedy**: The cached content is that the least popular will be replaced by the new content.
- 3) **Independent Q learning (IQL)**: The dynamic content replacement is operated by deploying the IQL without information interaction among agents [39].
- 4) **Traditional MARL**: This policy employs MARL framework without the low entropy communications [40].

Fig. 4 shows that the loss function of the proposed MARL-LEC policy and the traditional MARL policy varies with the time slots. It is observed that the loss gradually declines as time slots iterates. After certain time slots, the function converges to a stable value. By comparison, the proposed MARL-LEC policy may fluctuate at the initial stage of training. This is because the MARL-LEC policy needs to quantify messages and perform pseudo gradient operation of information entropy, resulting in slow convergence of the model. It can be seen that the traditional MARL policy can

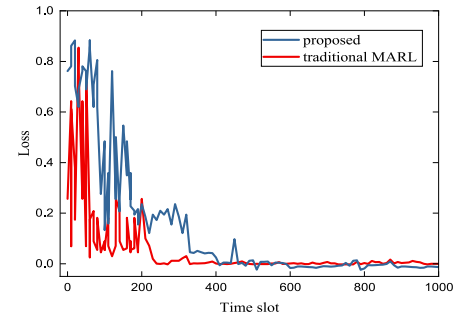


Fig. 4. Loss between current Q-values and target Q-values over time slots.

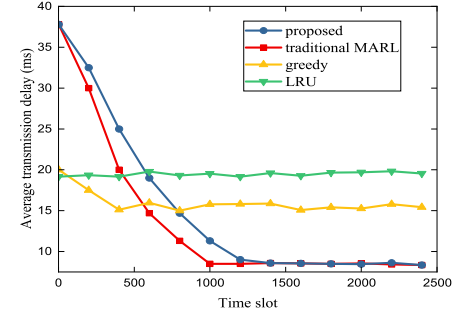


Fig. 5. Average transmission delay versus time slot for different caching policies.

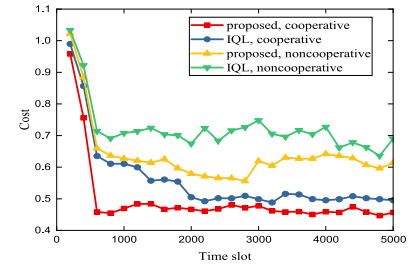


Fig. 6. Cost versus time slot for different caching policies.

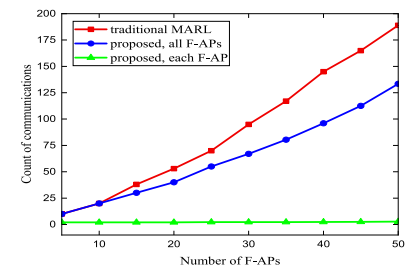


Fig. 7. Count of communication versus number of F-APs for different caching policies.

achieve stable convergence after 200 time slots, while the proposed MARL-LEC policy can achieve convergence after 400 time slots.

Fig. 5 shows that the average content transmission delay of the proposed MARL-LEC policy and three benchmark policies (traditional MARL, greedy and LRU) changes with the time slots. It can be observed that the content transmission delays of the traditional MARL policy and MARL-LEC policy are significantly higher than that of the LRU policy after the time slot $t < 500$. This is because the model has less data about

TABLE II
SUMMARY OF MAIN SIMULATION PARAMETERS

Parameter	Definition	Value
M	number of F-APs	20
U	number of user	500
F	number of content	1000
C_n	cache capacity of F-AP	200 MB
B	bandwidth	50 MHz
P_n	the transmit power of F-AP n	1 W
N_0	noise power spectral density	-174 dBm/Hz
η	the skewness factor	0.8
α	the learning rate of DDQN	0.001
γ	the discount factor of DDQN	0.9
$\lambda_1, \lambda_2, \lambda_3$	the weight factor of reward function	0.7, 0.2, 0.1
δ	the quantization interval length	0.25
β	the weight of the regularizer	0.1

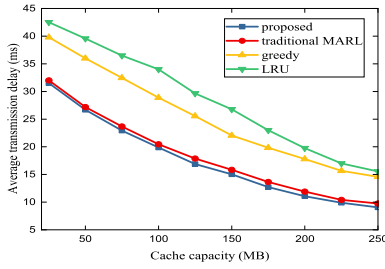


Fig. 8. Average transmission delay versus F-AP cache capacity for different caching policies.

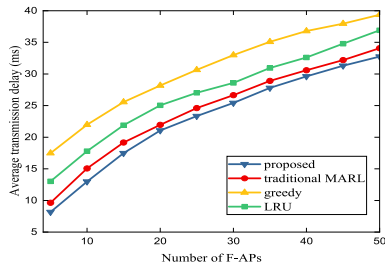


Fig. 9. Average transmission delay versus number of F-APs for different caching policies.

the state information at the initial training stage and the F-AP has a high decision delay. As the time slot increases, the content transmission delay of the traditional MARL policy and MARL-LEC policy decreases rapidly. After the time slot $t > 1500$, the content transmission delay of the traditional MARL policy and MARL-LEC policy is stable, and finally the optimal cache decision can be achieved.

Fig. 6 shows the normalized costs of the proposed MARL-LEC policy and IQL policy in both cooperative and non-cooperative caching scenarios as varying with time slots. The ratio of content transmission delay to the overall content transmission delay from the cloud server is known as normalized cost. It is clearly that the normalized costs of the four curves gradually converge to stable values as the training time increases. In addition, the non-cooperative scenarios have higher costs compared to cooperative scenarios,

and cooperative caching scenarios can bring the improvement in content transmission. This is due to the inability to share content between F-APs in a non-cooperative caching scenario, resulting in a large amount of content needing to be transmitted from cloud servers. By increasing the communication process between F-APs and enhancing their cooperation, the proposed MARL policy can have lower costs.

By setting the number of F-APs, we vary the number of cooperative F-APs in each F-AP $|\mathcal{N}_N|$. Fig. 7 elucidates the variation in the quantity of cooperative F-APs engaged in the communications process within the proposed MARL-LEC policy as compared to the traditional MARL policy, in relation to the total number of F-APs in the network. We observed that when the overall count of F-APs is relatively low, the proportion of cooperative F-APs in the communications process under the MARL-LEC policy mirrors that of the traditional MARL policy. However, as the total number of F-APs escalates, a notable divergence emerges. The frequency of cooperative F-APs in the communications process under the MARL-LEC policy tends to stabilize at approximately 2-3, in contrast to a continuous increase observed in the traditional MARL policy. This stabilization significantly curtails the extent of cooperation required among all F-APs in the network. Consequently, the MARL-LEC policy manifests a substantial reduction in communications instances, thereby markedly enhancing the overall communications efficiency within the network.

Fig. 8 shows that the average content transmission delay of the proposed MARL-LEC policy and the three benchmark policies (traditional MARL policy, greedy and LRU) changes with the cache capacity of the F-AP. The average content transmission delay for all four policies decreases as the cache capacity of the F-AP increases, which is clearly evident. This is because greater amount of popular contents can be locally cached under the larger cache capacity of F-AP, thereby minimizing the transmission delay. Additionally, compared to greedy policy and LRU policy, MARL-LEC policy and traditional MARL policy exhibit better cache performance.

Fig. 9 demonstrates the average transmission delay of the proposed cooperative caching policy compared with three benchmark policies (traditional MARL, greedy and LRU) as the number of F-APs. It is shown that the average transmission delay of four policies all increases as the number of F-APs increases. That is because the popularity of content varies among different F-APs, which has increased the varieties of content that users are requesting. Due to the unchanged cache capacity of F-APs, the transmission delay would increase. However, our proposed MARL-LEC policy can make the better caching strategy and deeply decrease the transmission delay compared with the benchmark policies.

Overall, simulation results clearly indicates the better cache performance by MARL-LEC policy under the different scenarios. Moreover, our proposed MARL-LEC policy can improve the communication efficiency without compromising the performance of cooperative caching.

V. CONCLUSION

In this paper, we have proposed the multi-agent reinforcement learning method with low entropy communication for cooperative edge caching in F-RANs. Considering the dynamic system environment, we have formulated the cooperative caching problem as an SG. We have employed DDQN in each F-AP to find the optimal caching policy by learning and interacting with the cooperative F-APs. We have also established an MARL framework to apply the communication process among F-APs. Each F-AP can interact with its cooperative F-APs and record the historical caching information as the belief of local update. In order to improve the communication efficiency, we quantize the message in the communication process and establish the low entropy communications. Finally, we have used the pseudo gradient method to solve the non-existent gradient of information entropy. We have theoretically analyzed the effectiveness of the pseudo gradient method. Simulation results have shown that our proposed policy can significantly improve the communication efficiency and achieve global convergence while maintaining the caching performance. One future work is to explore the model security and model transfer of the MARL framework in F-RANs.

APPENDIX A PROOF OF THEOREM 1

The parameter θ of all current networks can be expressed as:

$$\theta^{t+1} = \theta^t - \alpha \nabla F(\theta^t). \quad (35)$$

During each round of training, there should be $\max_n \|f_n(\theta_n^t)\| \leq B \|\nabla F(\theta^t)\|$, where B is a constant. For the optimization of loss function, there will be stagnation points at the same time.

The error defined during the iteration process is e^t , and Eq. (35) can be rewritten as:

$$\theta^{t+1} = \theta^t - \alpha (\nabla F(\theta^t) + e^t), \quad (36)$$

where $e^t = \nabla_n f_n(\theta_n^t) - \nabla F(\theta^t)$. Since the mean of error is equal to 0, it can be expressed as:

$$\begin{aligned} \mathbb{E}[e^t] &= \mathbb{E}[\nabla f_n(\theta_n^t) - \nabla F(\theta^t)] \\ &= \mathbb{E}[\nabla f_n(\theta_n^t)] - \nabla F(\theta^t) = 0. \end{aligned} \quad (37)$$

Furthermore, based on the derivations above, we have:

$$\begin{aligned} \mathbb{E}[\|e^t\|^2] &= \mathbb{E}[\|\nabla f_n(\theta_n^t) - \nabla F(\theta^t)\|^2] \\ &= \mathbb{E}[\|\nabla f_n(\theta_n^t)\|^2 - 2\langle \nabla f_n(\theta_n^t), \nabla F(\theta^t) \rangle \\ &\quad + \|\nabla F(\theta^t)\|^2] \\ &= \mathbb{E}[\|\nabla f_n(\theta_n^t)\|^2 - 2\langle \mathbb{E}[\nabla f_n(\theta_n^t)], \nabla F(\theta^t) \rangle \\ &\quad + \|\nabla F(\theta^t)\|^2] \\ &= \frac{1}{N} \sum_{n=1}^N [\|\nabla f_n(\theta_n^t)\|^2 - \|\nabla F(\theta^t)\|^2] \\ &\leq (B^2 - 1) \|\nabla F(\theta^t)\|^2. \end{aligned} \quad (38)$$

By combining inequalities (24) and Eq. (35), we can obtain:

$$F(\theta^{t+1}) \leq F(\theta^t) + \langle \nabla F(\theta^t), \theta^{t+1} - \theta^t \rangle + \frac{G}{2} \|\theta^{t+1} - \theta^t\|^2. \quad (39)$$

Then, by introducing the error e^t and Eq. (35) into the inequality (39), we can get:

$$\begin{aligned} F(\theta^{t+1}) &< F(\theta^t) - \alpha \langle \nabla F(\theta^t), \nabla F(\theta^t) + e^t \rangle \\ &\quad + \frac{\alpha^2 G}{2} \|\nabla F(\theta^t) + e^t\|^2 \\ &= F(\theta^t) - \alpha \left(1 - \frac{\alpha G}{2}\right) \|\nabla F(\theta^t)\|^2 \\ &\quad - \alpha(1 - \alpha G) \langle \nabla F(\theta^t), e^t \rangle + \frac{\alpha^2 G}{2} \|e^t\|^2. \end{aligned} \quad (40)$$

Due to the fact that the learning rate α can approach infinitesimal, the expectation is that the error e^t is taken on both sides of the inequality (40). Based on Eq. (37) and Eq. (38), new inequality can be expressed as:

$$\begin{aligned} \mathbb{E}[F(\theta^t)] &\leq F(\theta^t) - \alpha \left(1 - \frac{\alpha G}{2}\right) \|\nabla F(\theta^t)\|^2 \\ &\quad - \alpha(1 - \alpha G) \langle \nabla F(\theta^t), \mathbb{E}[e^t] \rangle \\ &\quad + \frac{\alpha^2 G(B^2 - 1)}{2} \|\nabla F(\theta^t)\|^2 \\ &\leq F(\theta^t) - \alpha \left(1 - \frac{\alpha G}{2}\right) \|\nabla F(\theta^t)\|^2 \\ &\quad + \frac{\alpha^2 G(B^2 - 1)}{2} \|\nabla F(\theta^t)\|^2 \\ &\leq F(\theta^t) - \alpha \left(1 - \frac{\alpha G B^2}{2}\right) \|\nabla F(\theta^t)\|^2. \end{aligned} \quad (41)$$

The inequality (41) indicates that when the learning rate is small enough and satisfies $0 < \alpha < \frac{2}{GB^2}$, SGD can obtain the expected parameter θ , and the corresponding state value function can be updated based on the obtained parameter θ . The convergence proof in MARL policy is completed.

APPENDIX B

PROOF OF THEOREM 2

Set $G_k = \epsilon + \sum_{n=1}^N I_k(m_n)$. G_K indicates the number of message variables located over the quantization interval $[(k-0.5)\Delta - 1, (k+0.5)\Delta - 1)$. To prove Theorem 2, we need three lemmas.

Lemma 1: If $m_n \in (u\Delta - 1, (u+1)\Delta - 1)$, then:

$$\begin{aligned} \nabla_{m_n}^p H(\mathbf{M}) &> 0, & \text{if } N_u > N_{u+1} \\ \nabla_{m_n}^p H(\mathbf{M}) &< 0, & \text{if } N_u < N_{u+1} \\ \nabla_{m_n}^p H(\mathbf{M}) &= 0, & \text{if } N_u = N_{u+1}. \end{aligned} \quad (42)$$

This indicates that for two adjacent quantization intervals, the pseudo gradient descent strategy moves the message variable from unpopular to popular.

Proof: If $m_n \in (u\Delta - 1, (u+1)\Delta - 1)$, we have:

$$\begin{aligned} \nabla_{m_n}^p H(\mathbf{M}) &= \sum_{k=0}^K \nabla_{I_k} H(\mathbf{M}) p_k(m_n) \\ &= -\frac{1}{N} \sum_{k=0}^K (1 + \log \frac{N_k}{N}) p_k(m_n). \end{aligned} \quad (43)$$

For $\forall m_n \in (u\Delta - 1, (u+1)\Delta - 1)$, we all have $s_u(m_n) = -1, s_{u+1} = 1$. And for $\forall k \in \{0, 1, \dots, K\} \setminus \{u, u+1\}$, we have $p_k(m_n) = 0$. Therefore, we have:

$$\nabla_{m_n}^p H(\mathbf{M}) = -\frac{1}{N} \log \frac{N_{u+1}}{N_u}. \quad (44)$$

According to Eq. (44), it can be proven that the Lemma 1 holds.

Lemma 2: If $N_u > N_{u+1}$, the update rule of $m_n \in (u\Delta - 1, (u+1)\Delta - 1)$ is given by:

$$\eta \rightarrow 0, \quad m'_n = m_n - \eta \nabla_{m_n}^p H(\mathbf{M}). \quad (45)$$

This update rule yields only two results:

- 1) $N'_k = N_k, \forall k \in \{0, 1, \dots, K\}$;
- 2) $N'_u = N_u + 1, N'_{u+1} = N_{u+1} - 1, N'_k = N_k, \forall k \in \{0, 1, \dots, K\} \setminus \{u, u+1\}$.

Similarly, if $N_u < N_{u+1}$, the update rule in Eq. (45) for $m_n \in (u\Delta - 1, (u+1)\Delta - 1)$ also yields only two results:

- 1) $N'_k = N_k, \forall k \in \{0, 1, \dots, K\}$;
- 2) $N'_u = N_u - 1, N'_{u+1} = N_{u+1} + 1, N'_k = N_k, \forall k \in \{0, 1, \dots, K\} \setminus \{u, u+1\}$.

Proof: First, consider the scenario $N_u > N_{u+1}$. For $\nabla_{m_n}^p H(\mathbf{M}) > 0$, we have

$$\begin{aligned} m'_n &= m_n - \eta \nabla_{m_n}^p H(\mathbf{M}) \in \\ &\quad \times [m_n - \eta \max(\nabla_{m_n}^p H(\mathbf{M})), m_n], \end{aligned} \quad (46)$$

where $\max(\nabla_{m_n}^p H(\mathbf{M}))$ can be expressed as:

$$\max(\nabla_{m_n}^p H(\mathbf{M})) = \frac{1}{N} \log \frac{N_s + \epsilon}{\epsilon}. \quad (47)$$

When $\eta \rightarrow 0$, we have $\eta \max(\nabla_{m_n}^p H(\mathbf{M})) < 0.5\Delta$. Then for $m_n \in (u\Delta - 1, (u+1)\Delta - 1)$, the range of m'_n under the updating of Eq. (46) can be $m'_n \in ((u-0.5)\Delta - 1, (u+1)\Delta - 1)$. Next, there are two scenarios for different values of m_n and m'_n :

1) If $m_n \in ((u+0.5)\Delta - 1, (u+1)\Delta - 1)$ and $m'_n \in ((u-0.5)\Delta - 1, (u+0.5)\Delta - 1)$, then we have $N'_u = N_u + 1, N'_{u+1} = N_{u+1} - 1, N'_k = N_k, \forall k \in \{0, 1, \dots, K\} \setminus \{u, u+1\}$.

2) Otherwise, we have $N'_k = N_k, \forall k \in \{0, 1, \dots, K\}$.

Similarly, the proof of scenario $N_u < N_{u+1}$ can be carried out. Lemma 2 is now proven.

Lemma 3: Let $H(\mathbf{M})$ represent $H(m_1, \dots, m_n, \dots, m_N)$ and $H(\mathbf{M}')$ express $H(m_1, \dots, m'_n, \dots, m_N)$.

If $N_u > N_{u+1}, N'_u = N_u + 1, N'_{u+1} = N_{u+1} - 1, N'_k = N_k, \forall k \in \{0, 1, \dots, K\} \setminus \{u, u+1\}$ or $N_u < N_{u+1}, N'_u = N_u - 1, N'_{u+1} = N_{u+1} + 1, N'_k = N_k, \forall k \in \{0, 1, \dots, K\} \setminus \{u, u+1\}$, we have $H(\mathbf{M}') < H(\mathbf{M})$.

Proof: First, consider the first scenario $N_u > N_{u+1}$:

$$\begin{aligned} H(\mathbf{M}') - H(\mathbf{M}) &= \frac{N_u}{N} \log \frac{N_u}{N} + \frac{N_{u+1}}{N} \log \frac{N_{u+1}}{N} \\ &\quad - \frac{N_u + 1}{N} \log \frac{N_u + 1}{N} \\ &\quad - \frac{N_{u+1} - 1}{N} \log \frac{N_{u+1} - 1}{N}. \end{aligned} \quad (48)$$

According to the mean value theorem, there exists $x_1 \in (\frac{N_u}{N}, \frac{N_{u+1}}{N})$ and $x_2 \in (\frac{N_{u+1}-1}{N}, \frac{N_{u+1}}{N})$ satisfying:

$$\begin{aligned} \frac{\log x_1 + 1}{N} &= \frac{N_u + 1}{N} \log \frac{N_u + 1}{N} - \frac{N_u}{N} \log \frac{N_u}{N}, \\ \frac{\log x_2 + 1}{N} &= \frac{N_{u+1}}{N} \log \frac{N_{u+1}}{N} - \frac{N_{u+1} - 1}{N} \log \frac{N_{u+1} - 1}{N}. \end{aligned} \quad (49)$$

According to Eq. (48) and Eq. (49), we have:

$$H(\mathbf{M}') - H(\mathbf{M}) = \frac{\log x_2 + 1}{N} - \frac{\log x_1 + 1}{N}. \quad (50)$$

Since $x_2 < \frac{N_{u+1}}{N} < \frac{N_u}{N} < x_1$, we have $H(\mathbf{M}') < H(\mathbf{M})$.

Next, consider the second scenario $N_u < N_{u+1}$:

$$\begin{aligned} H(\mathbf{M}') - H(\mathbf{M}) &= \frac{N_u}{N} \log \frac{N_u}{N} + \frac{N_{u+1}}{N} \log \frac{N_{u+1}}{N} \\ &\quad - \frac{N_u - 1}{N} \log \frac{N_u - 1}{N} \\ &\quad - \frac{N_{u+1} + 1}{N} \log \frac{N_{u+1} + 1}{N}. \end{aligned} \quad (51)$$

Similarly, according to the mean value theorem, there exist $x_3 \in (\frac{N_u-1}{N}, \frac{N_u}{N})$ and $x_4 \in (\frac{N_{u+1}}{N}, \frac{N_{u+1}+1}{N})$ satisfying:

$$\begin{aligned} \frac{\log x_3 + 1}{N} &= \frac{N_u}{N} \log \frac{N_u}{N} - \frac{N_u - 1}{N} \log \frac{N_u - 1}{N} \\ \frac{\log x_4 + 1}{N} &= \frac{N_{u+1} + 1}{N} \log \frac{N_{u+1} + 1}{N} - \frac{N_{u+1}}{N} \log \frac{N_{u+1}}{N}. \end{aligned} \quad (52)$$

According to Eq. (51) and Eq. (52), we have:

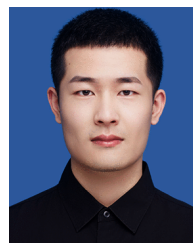
$$H(\mathbf{M}') - H(\mathbf{M}) = \frac{\log x_3 + 1}{N} - \frac{\log x_4 + 1}{N}. \quad (53)$$

Since $x_3 < \frac{N_{u+1}}{N} < \frac{N_u}{N} < x_4$, we have $H(\mathbf{M}') < H(\mathbf{M})$. Lemma 3 is now proven.

According to Lemma 2 and Lemma 3, if $H(\mathbf{M}') \neq H(\mathbf{M})$, we have $H(\mathbf{M}') < H(\mathbf{M})$. Therefore, $H(\mathbf{M}') \leq H(\mathbf{M})$ can be obtained, which completes the proof.

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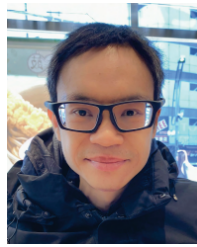
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